

Implementation / Feasibility Plan

How ASTRA goes from a working model to use across the state

No sudden switchover. ASTRA earns trust one step at a time.

It starts by only watching, then starts helping, and finally runs the backup and recovery, alongside the exam companies already in use.

Can it be done?

Yes

Technically: uses only proven, free software and standard security methods

Yes

Practically: adds just 2 small servers, a battery-backup check and clear instructions per centre

Yes

Financially: no licence fees; modest hardware that can be reused

Needs rules

Policy: the exam authority must set limits, e.g. maximum pause time

Step-by-step roll-out

Phase	Time	What happens	Move on when
0 • Working model	1–2 months	Tested on simulated exam centres	Every type of failure recovers with no answers lost
1 • Watch only	2–3 months	Runs at 3–5 real centres during real exams, only watching ; it doesn't touch the exam	The live view is accurate; AI predictions are checked against real problems
2 • Assist	3 months	Instructions, candidate messages and decision help switched on; answer backup runs alongside the current system	Problems handled faster; records match the current system
3 • Full trial	3 months	One complete exam uses ASTRA for backup and recovery	No lost answers; no needless re-exams
4 • State-wide	6–12 months	All centres of selected exam bodies	Fewer cancellations than in past years
5 • Expand	Ongoing	Other exam boards, other states, and online answer-sheet marking	–

What each centre needs



Current exam PCs

With the locked-down exam setup, or just a small monitoring program in the early phases



2 small servers

One main and one spare, each with its own battery backup, on different power lines



Network

Separate networks for exam PCs and cameras; a secure internet link, with a 4G backup



CCTV

The centre's existing cameras and recorder



Invigilator device

Any tablet or PC with a web browser



Control room

A main and a backup site in a government data centre

Working with current systems and people

- **Works next to current exam companies:** a small program collects health signals, and connectors read exam events. It starts in "watch only" mode, and a standard format makes it easy for any vendor to connect.
- **Training:** invigilators 1–2 hours; centre staff 1 day; exam officials 1 day; auditors half a day.
- **A backup drill before every exam** (switch off the main server, check the spare takes over) doubles as practical training.

Rules and legal compliance



Decided by the exam authority

Maximum pause time, how lost time is returned, who approves each decision step, what happens after a long pause, whether the internet link stays on during the exam, and how long CCTV footage is kept



Legal compliance

Data Protection Act 2023: minimal personal data, access records, CCTV notices. Public Examinations Act 2024: usable evidence. An independent, government-approved (CERT-In empanelled) security audit before the full trial.

Risks and how we handle them

Risk	What we do
Centres with poor power backup	The pre-exam health check shows this early; the centre server's own battery protects the answers
Exam companies unwilling to connect	Start in watch-only mode; a standard connection format; add it to future contracts
Staff new to the system	Short training, practice drills, one simple dashboard
AI makes mistakes early on	Officials approve important actions; simple rules as backup; AI retrained on real data
Staff misusing access	The "blind courier" server, two-person approval, every action recorded
Very large exams	Each centre runs on its own; the control room receives only summaries; traffic is controlled

Free software, no lock-in

Reuses existing PCs, networks and CCTV

Gets smarter after every incident

Can extend to online marking